159102 workshop – Week 3

1. What is the hexadecimal of the bit pattern for 32-bit storage of the floating point numbers:

a) 100.875 **b)** -0.75

2. What floating point numbers are stored (32 bits) as the patterns with hexadecimal:

a) 42590000 **b)** BF900000

3. Working in **4 bits**, give answers in binary, hexadecimal and decimal to:

a) 6 & 3 **b)** 9 ^ 5

4. Working in **8 bits**, give answers in binary, hexadecimal and decimal to:

a) C3 | 5F **b)** DE << 3

5. Assuming unsigned integer storage:

a) show that a shift right achieves division by 2, by doing 12 >> n (in 4 bits) for n = 1, 2, and 3

b) show that a shift left achieves multiplication by 2, by doing 1 << n (in 4 bits) for n = 1, 2, and 3

6. Explain the difference in the results when using the following code:

```
c = 0xA4;
c = c >> 2;
cout << "c is " << (int) c << endl;
```

a) define `c` as a `char`

b) define `c` as an `unsigned char`

7. `c` is an `unsigned char`. Write code that uses bitwise operations to find the value in:

a) the least significant bit

b) the most significant bit

c) the 3 bits on the right (the low end of the number)

d) the 3 bits on the left (the top end of the number)